

Splunk SIEM SOC Lab#

Lab Introduction & Hands-On

Intoduction:

1. Objective:

Build a Splunk-based SIEM lab for SOC Analysis using Windows/DNS logs

2. Lab Architecture:

- Host OS: Kali Linux
- Guest OS: Windows 11 VM
- SIEM: Splunk Enterprise
- Forwarder: Splunk Universal Forwarder

Log Sources:

- Windows Security Events
- System Monitor (Sysmon)
- DNS/HTTP

3. Tools Employed

Tool	Purpose
Splunk Enterprise	SIEM
Splunk UF	Log forwarding
Sysmon	Endpoint telemetry

4. Data Onboarding:

- WinEventLog:Security
- Sysmon Operational Log
- DNS/HTTP Log

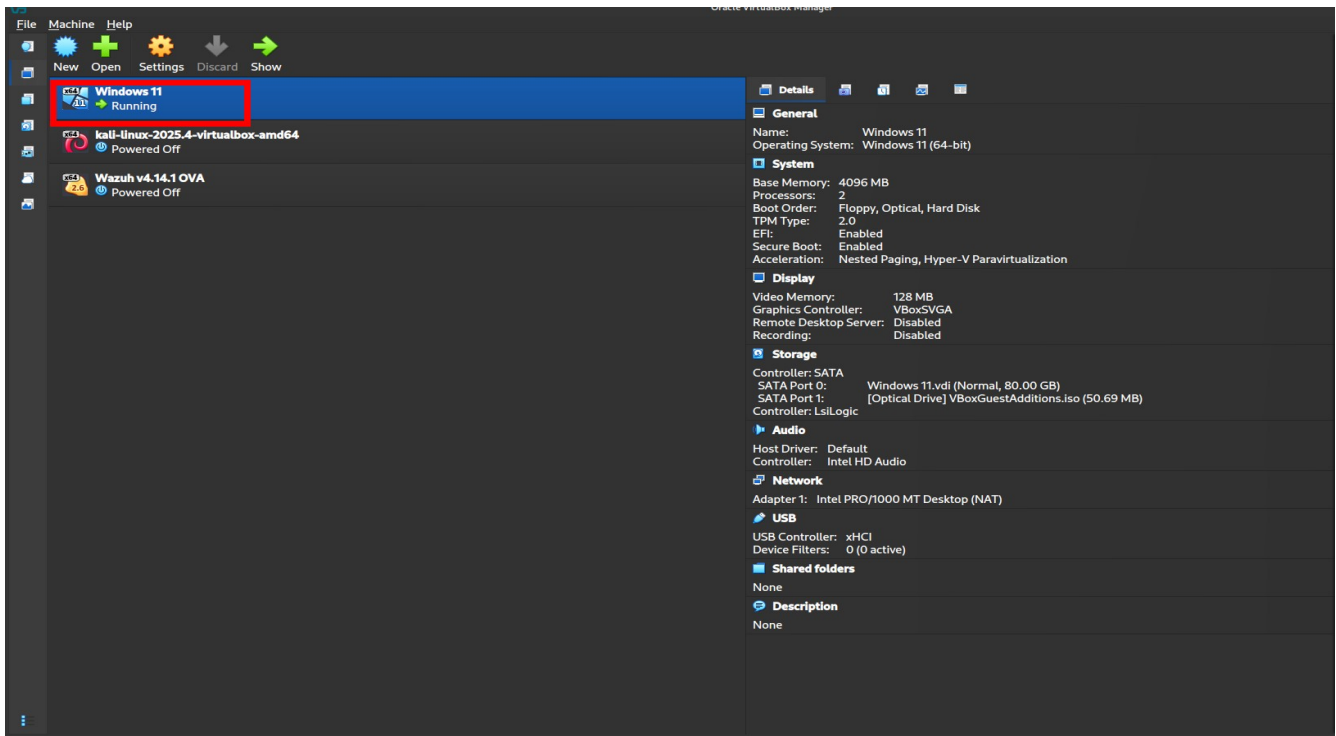
Hands-On:

A. Splunk Universal Forwarder Installation/Setup:

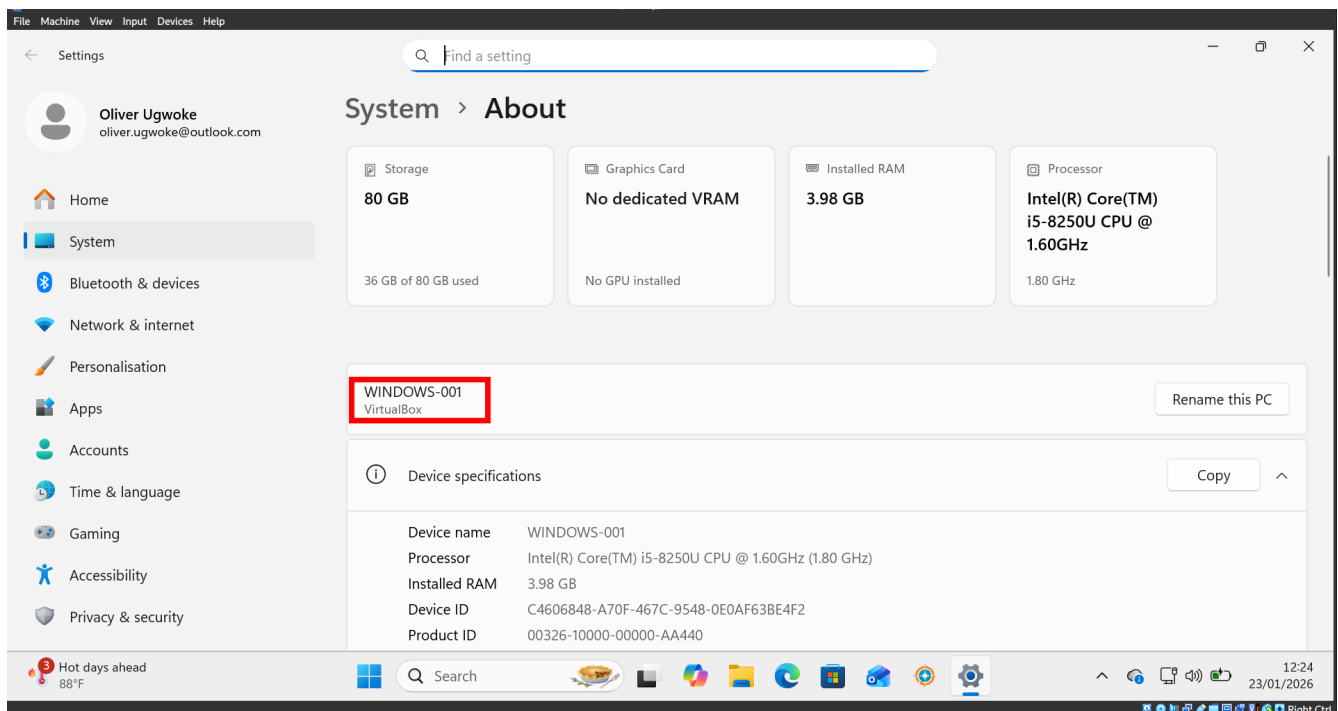
- “UF” installation on Windows VM and “inputs.conf” configuration

Step 1:

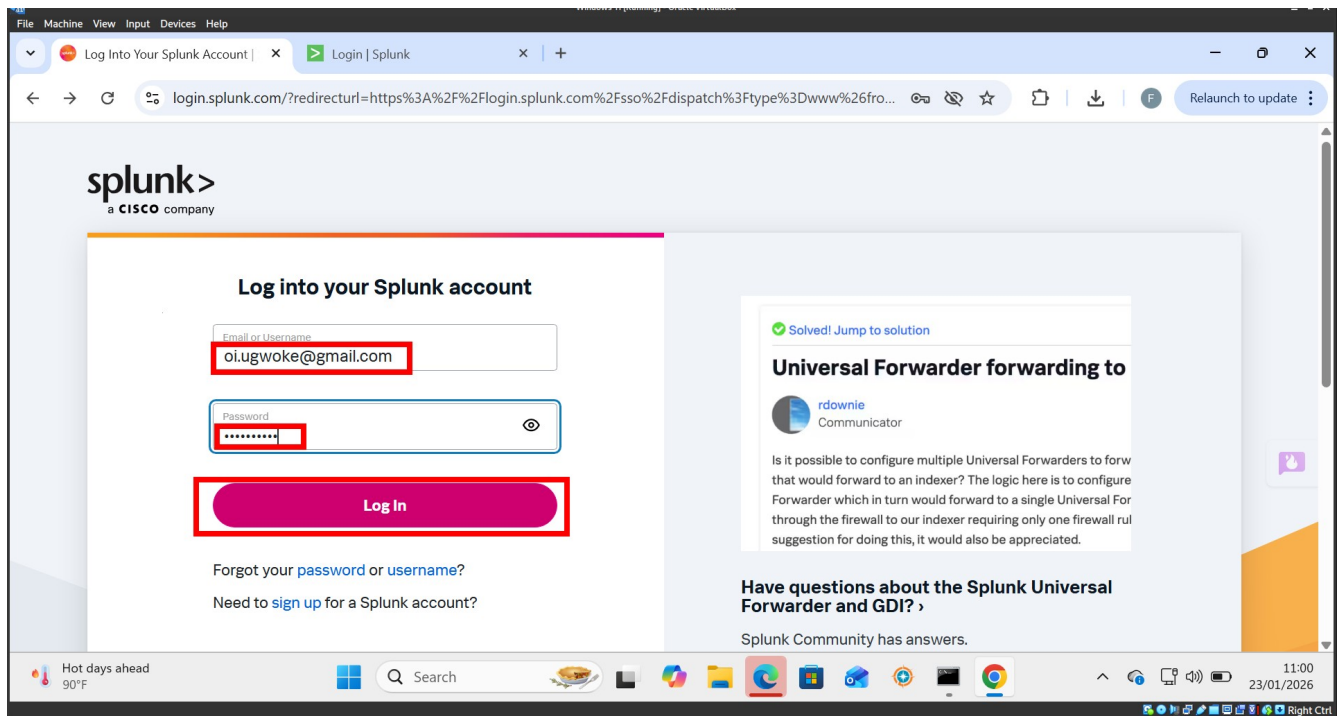
Guest OS (Windows 11)



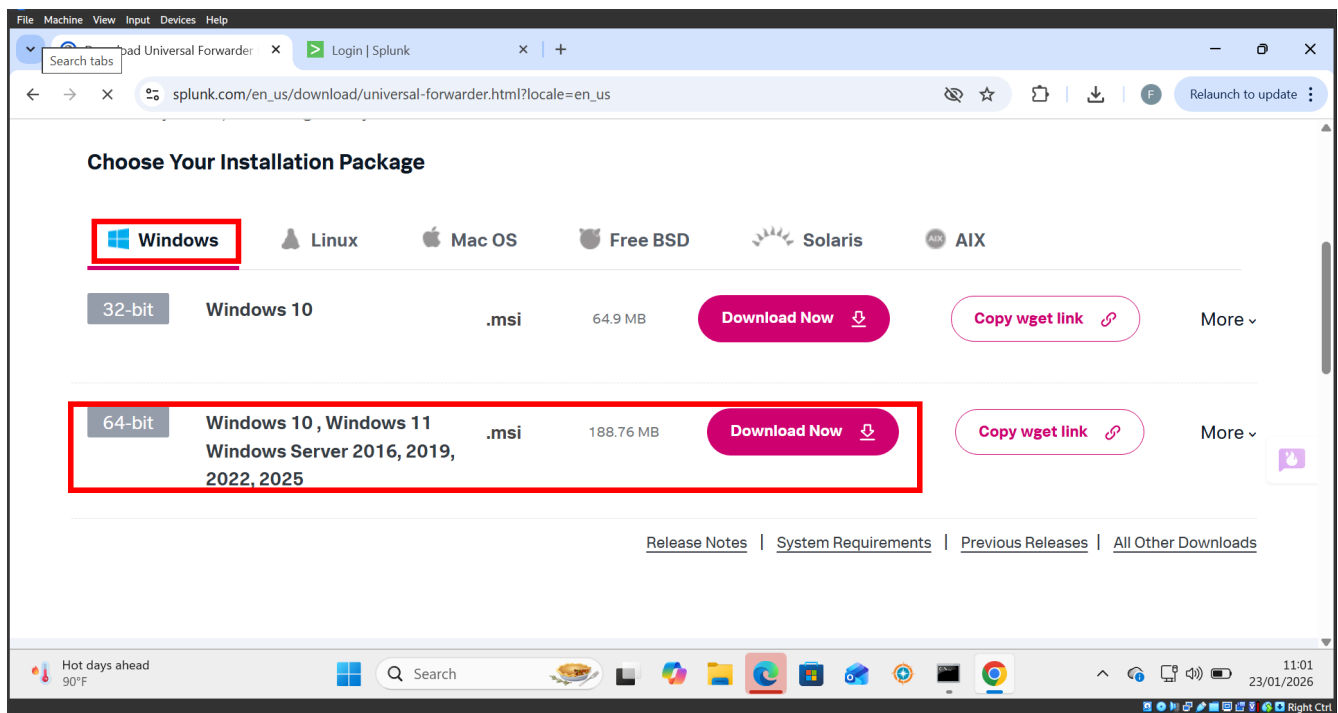
Step 2: Guest (Windows Name)



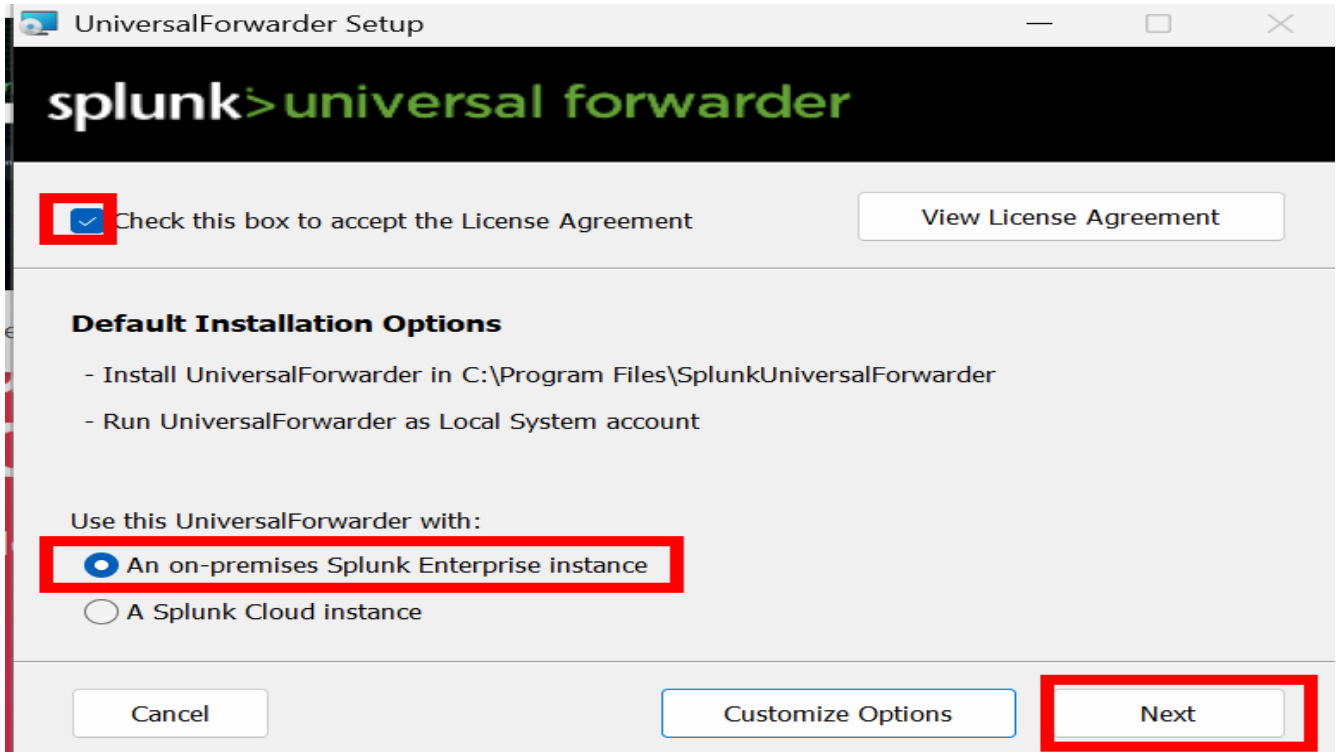
Step 3: Splunk Login



Step 4: UF Download



Step 5:
Check the selected boxes below and click “next”



The image shows the 'UniversalForwarder Setup' window. At the top is the Splunk logo and the text 'universal forwarder'. Below this, there is a checkbox labeled 'Check this box to accept the License Agreement' which is checked and highlighted with a red box. To its right is a button labeled 'View License Agreement'. Below this section is a heading 'Default Installation Options' followed by two bullet points: '- Install UniversalForwarder in C:\Program Files\SplunkUniversalForwarder' and '- Run UniversalForwarder as Local System account'. Underneath is the text 'Use this UniversalForwarder with:' followed by two radio button options: 'An on-premises Splunk Enterprise instance' (which is selected and highlighted with a red box) and 'A Splunk Cloud instance'. At the bottom of the window are three buttons: 'Cancel', 'Customize Options', and 'Next' (which is highlighted with a red box).

UniversalForwarder Setup

splunk>universal forwarder

☒ Check this box to accept the License Agreement [View License Agreement](#)

Default Installation Options

- Install UniversalForwarder in C:\Program Files\SplunkUniversalForwarder
- Run UniversalForwarder as Local System account

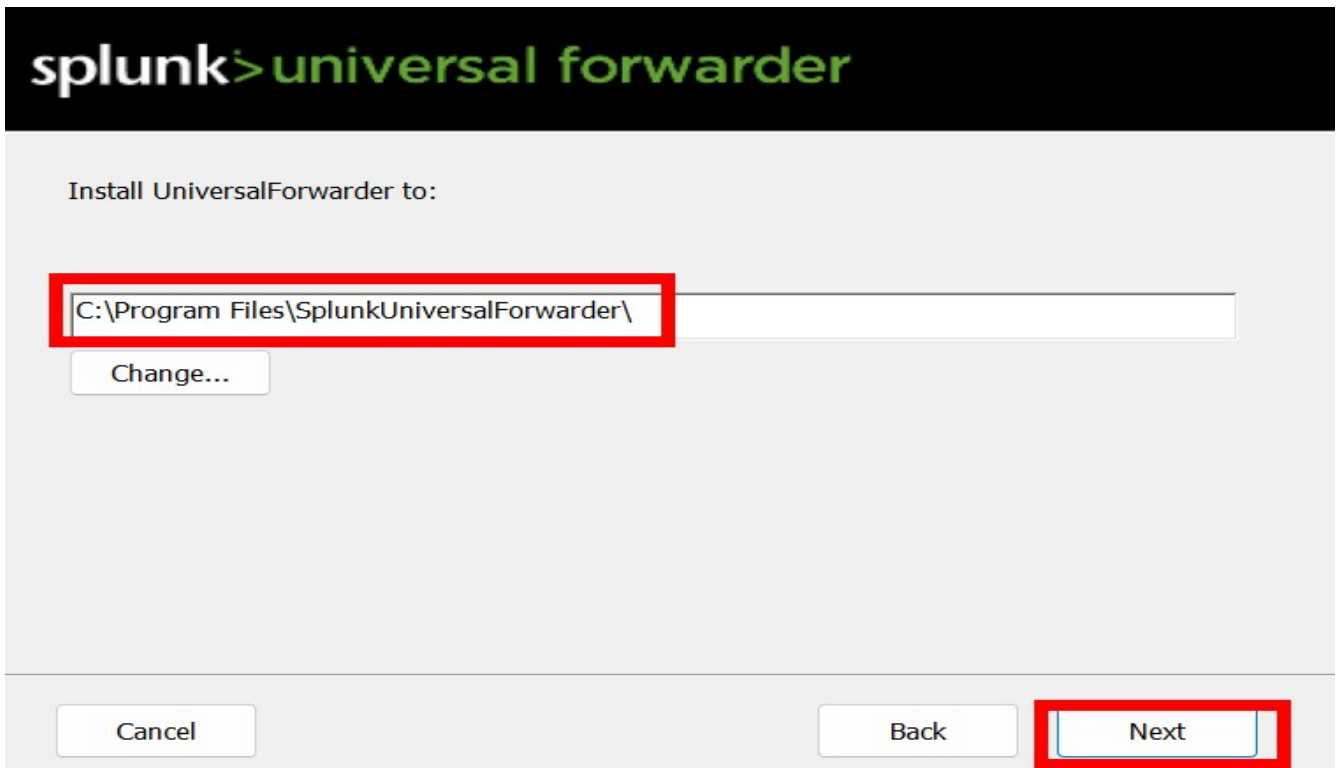
Use this UniversalForwarder with:

☒ An on-premises Splunk Enterprise instance

☐ A Splunk Cloud instance

[Cancel](#) [Customize Options](#) [Next](#)

Step 6:
Click “Next”



The image shows the 'UniversalForwarder Setup' window at a later stage. The top section with the Splunk logo and 'universal forwarder' text is still present. Below this, the text 'Install UniversalForwarder to:' is followed by a text input field containing the path 'C:\Program Files\SplunkUniversalForwarder\' which is highlighted with a red box. Below the input field is a button labeled 'Change...'. At the bottom of the window are three buttons: 'Cancel', 'Back', and 'Next' (which is highlighted with a red box).

splunk>universal forwarder

Install UniversalForwarder to:

C:\Program Files\SplunkUniversalForwarder\

[Change...](#)

[Cancel](#) [Back](#) [Next](#)

Step 7:
Click “Next”

The screenshot shows the 'splunk>universal forwarder' configuration window. At the top, a black header contains the text 'splunk>universal forwarder' in white and green. Below the header, a message states: 'If the following information is not provided, forwarded Splunk data will still be encrypted with the default Splunk certificate'. The form includes three main sections: 1. 'SSL certificate (file containing public and private key parts)' with a text input field and a 'Browse...' button. 2. 'Certificate Password' with 'Password:' and 'Confirm password:' labels, each followed by a text input field. 3. 'SSL root CA (the file containing the Root CA certificate to validate the server certificate)' with a text input field and a 'Browse...' button. At the bottom, there are three buttons: 'Cancel', 'Back', and 'Next'. The 'Next' button is highlighted with a red rectangular border.

Step 8:
Select “Local System” and click “Next”

The screenshot shows the 'splunk>universal forwarder' configuration window. At the top, a black header contains the text 'splunk>universal forwarder' in white and green. Below the header, a message states: 'The user you install UniversalForwarder as determines what data it has access to. The Managed Service Account and Group-Managed Service Account are supported by CLI only.' The form is titled 'Install UniversalForwarder as:' and lists three options: 1. 'Local System' with an unselected radio button, highlighted by a red rectangular border. 2. 'Domain Account' with an unselected radio button. 3. 'Virtual Account' with a selected radio button (indicated by a blue dot). Each option is followed by a descriptive paragraph. At the bottom, there are three buttons: 'Cancel', 'Back', and 'Next'. The 'Next' button is highlighted with a red rectangular border.

Step 9:

Check the below boxes to grant windows privileges to enable UF features and click “Next”

The screenshot shows the 'splunk>universal forwarder' configuration window. It has a dark header with the title in green. The main area is light gray and contains two sections of checkboxes. The first section, 'Grant Windows privileges to enable Universal Forwarder features', has three items: 'SeBackupPrivilege' (checked), 'SeSecurityPrivilege' (checked), and 'SeImpersonatePrivilege' (unchecked). The second section, 'Grant Windows groups privileges to enable Universal Forwarder features', has one item: 'Performance Monitor Users' (checked). A 'More Info' button is on the right. At the bottom are 'Cancel', 'Back', and 'Next' buttons, with the 'Next' button highlighted by a red rectangle.

splunk>universal forwarder

Grant Windows privileges to enable Universal Forwarder features

- ☒ **SeBackupPrivilege**
Required by "file monitor" inputs to bypass file permission checks
- ☒ **SeSecurityPrivilege**
Required by WinEventLog inputs to pull security event logs
- ☐ **SeImpersonatePrivilege**
Enables collecting events as a specific user/group

Grant Windows groups privileges to enable Universal Forwarder features

- ☒ **Performance Monitor Users**
Required by WMI/perfmon inputs to collect performance data

[More Info](#)

[Cancel](#) [Back](#) [Next](#)

Step 10:

Check all the boxes to allow UF to forwards all logs and click “Next”

The screenshot shows the 'splunk>universal forwarder' configuration window. It has a dark header with the title in green. The main area is light gray and contains two columns of checkboxes. The left column, 'Windows Event Logs', has five items: 'Application Logs', 'Security Log', 'System Log', 'Forwarded Events Log', and 'Setup Log', all checked. Below this is 'Active Directory Monitoring' with 'Enable AD monitoring' checked. The right column, 'Performance Monitor', has four items: 'CPU Load', 'Memory', 'Disk Space', and 'Network Stats', all checked. At the bottom is a 'Path to monitor' text box and two buttons: 'File...' and 'Directory...'. At the very bottom are 'Cancel', 'Back', and 'Next' buttons, with the 'Next' button highlighted by a red rectangle.

splunk>universal forwarder

Windows Event Logs

- ☒ Application Logs
- ☒ Security Log
- ☒ System Log
- ☒ Forwarded Events Log
- ☒ Setup Log

Active Directory Monitoring

- ☒ Enable AD monitoring

Performance Monitor

- ☒ CPU Load
- ☒ Memory
- ☒ Disk Space
- ☒ Network Stats

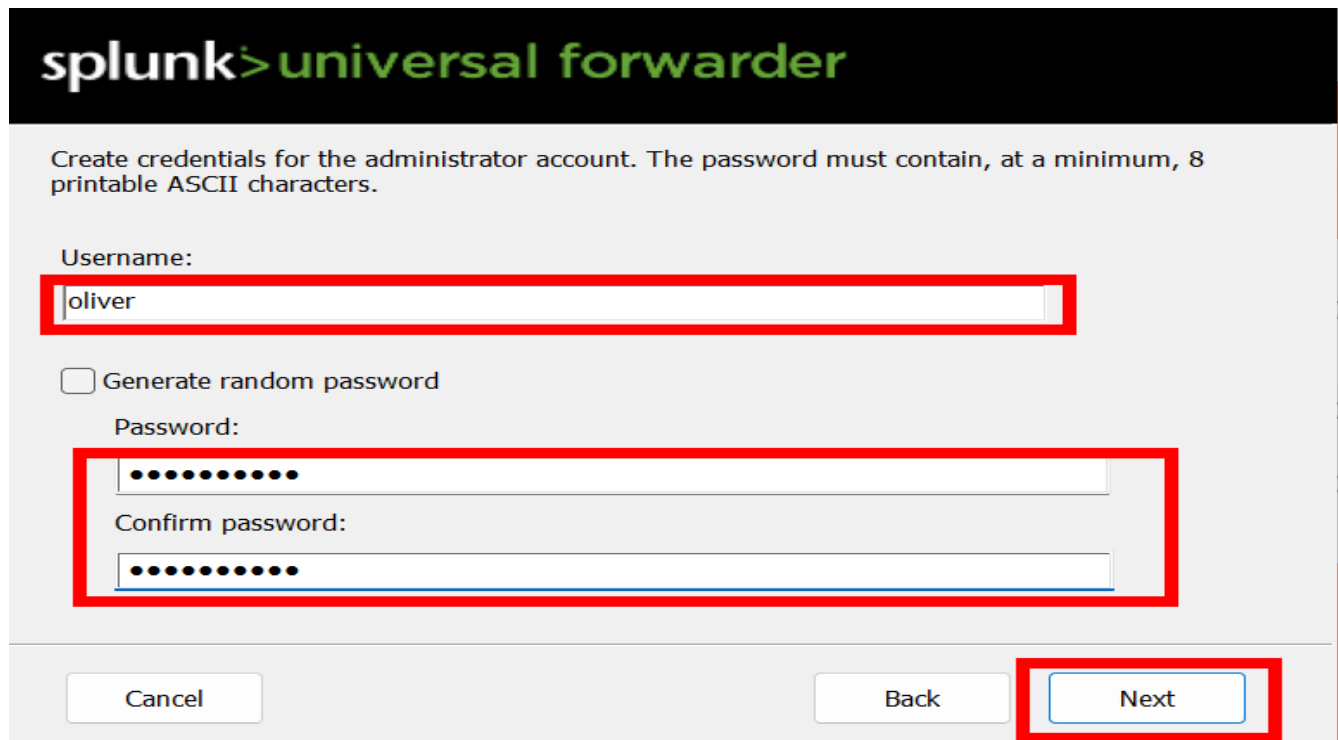
Path to monitor

[File...](#) [Directory...](#)

[Cancel](#) [Back](#) [Next](#)

Step 11:

Uncheck “Generate random Psw”, enter credentials to create administrator’s account and click “Next”



The screenshot shows the 'splunk>universal forwarder' setup window. The title bar is black with the text 'splunk>universal forwarder' in white and green. Below the title bar, a message states: 'Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.' The 'Username:' field contains 'oliver'. The 'Generate random password' checkbox is unchecked. The 'Password:' and 'Confirm password:' fields both contain masked text (dots). At the bottom, there are three buttons: 'Cancel', 'Back', and 'Next'. The 'Next' button is highlighted with a red border.

splunk>universal forwarder

Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.

Username:

oliver

☐ Generate random password

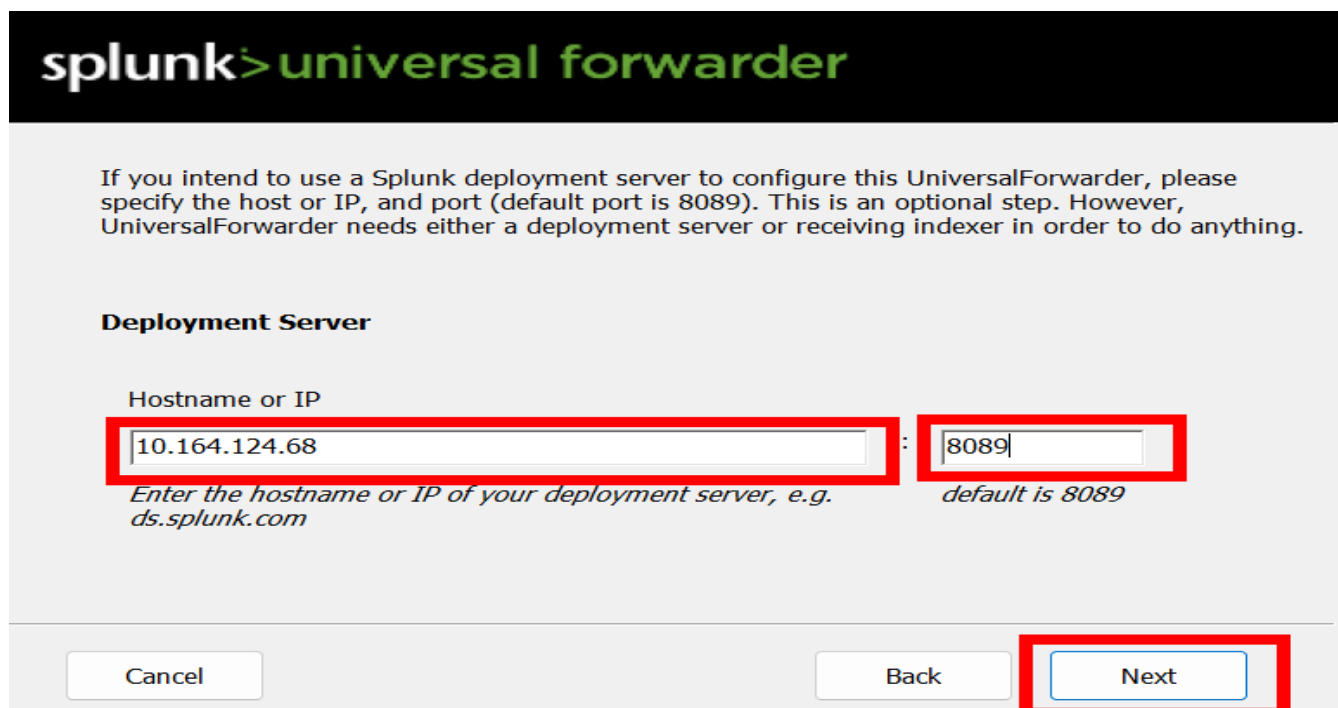
Password:

Confirm password:

Cancel Back Next

Step 12:

Enter host IP (Where Splunk Enterprise sits), default port “8089” to allow UF perform and click “Next”



The screenshot shows the 'splunk>universal forwarder' setup window. The title bar is black with the text 'splunk>universal forwarder' in white and green. Below the title bar, a message states: 'If you intend to use a Splunk deployment server to configure this UniversalForwarder, please specify the host or IP, and port (default port is 8089). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.' The 'Deployment Server' section has a 'Hostname or IP' field containing '10.164.124.68' and a port field containing '8089'. Below the fields, there is a note: 'Enter the hostname or IP of your deployment server, e.g. ds.splunk.com' and 'default is 8089'. At the bottom, there are three buttons: 'Cancel', 'Back', and 'Next'. The 'Next' button is highlighted with a red border.

splunk>universal forwarder

If you intend to use a Splunk deployment server to configure this UniversalForwarder, please specify the host or IP, and port (default port is 8089). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.

Deployment Server

Hostname or IP

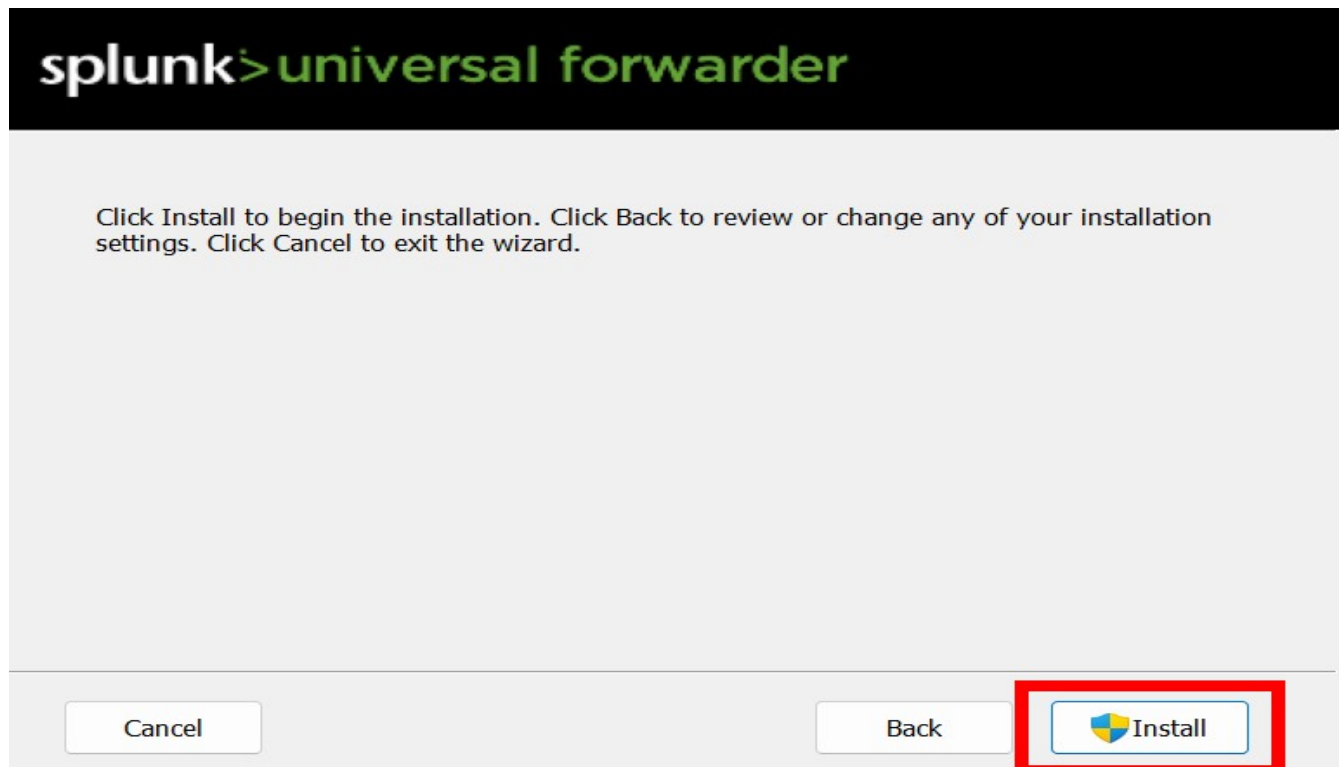
10.164.124.68 : 8089

Enter the hostname or IP of your deployment server, e.g. ds.splunk.com default is 8089

Cancel Back Next

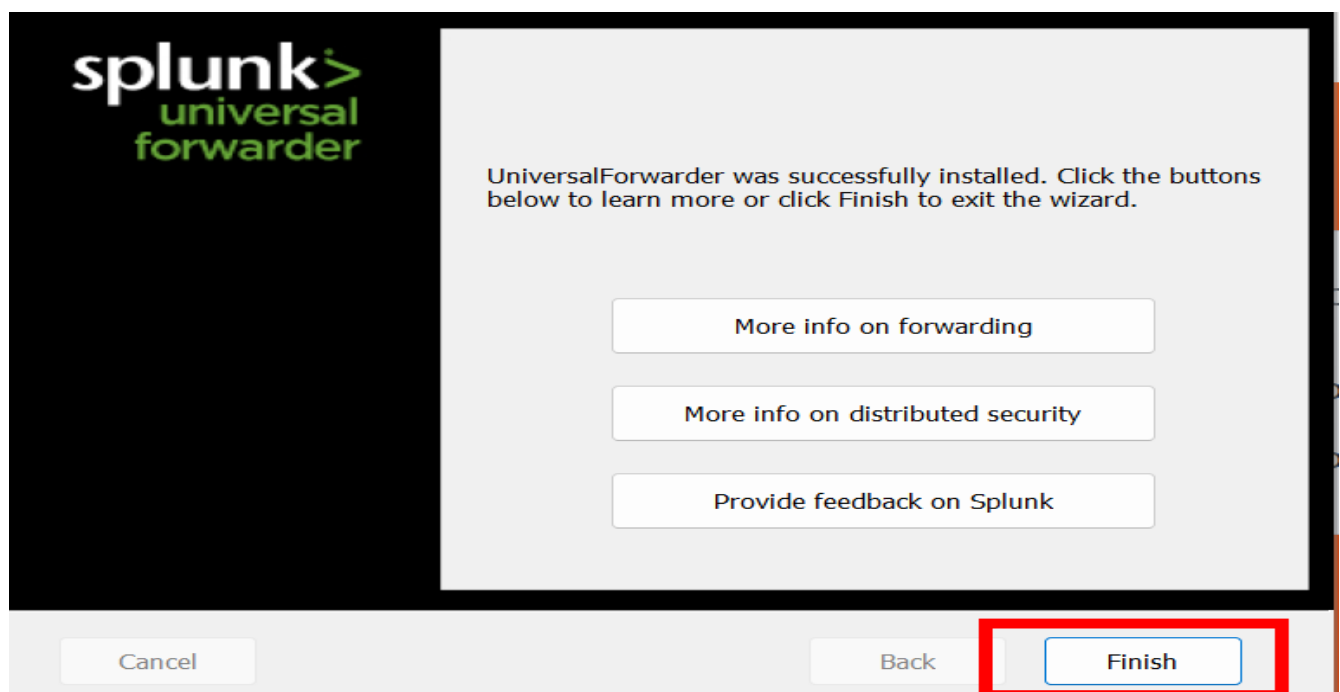
Step 13:

Click “Install” after configuring Splunk Enterprise instance to receive data from UF on “9997”



Step 14:

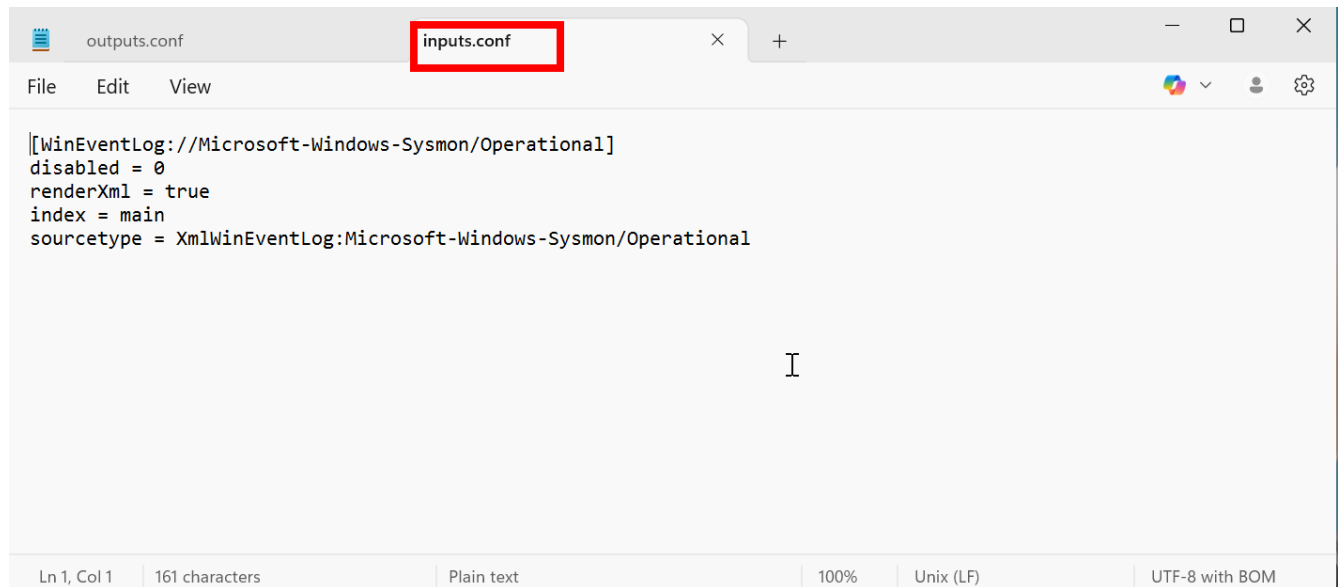
Click “Finish” to conclude UF installation



Step 15:

“Input.conf” configuration -

C: \Program File\SplunkUniversalForwarder\apps\TA-Sysmon\local\inputs.conf

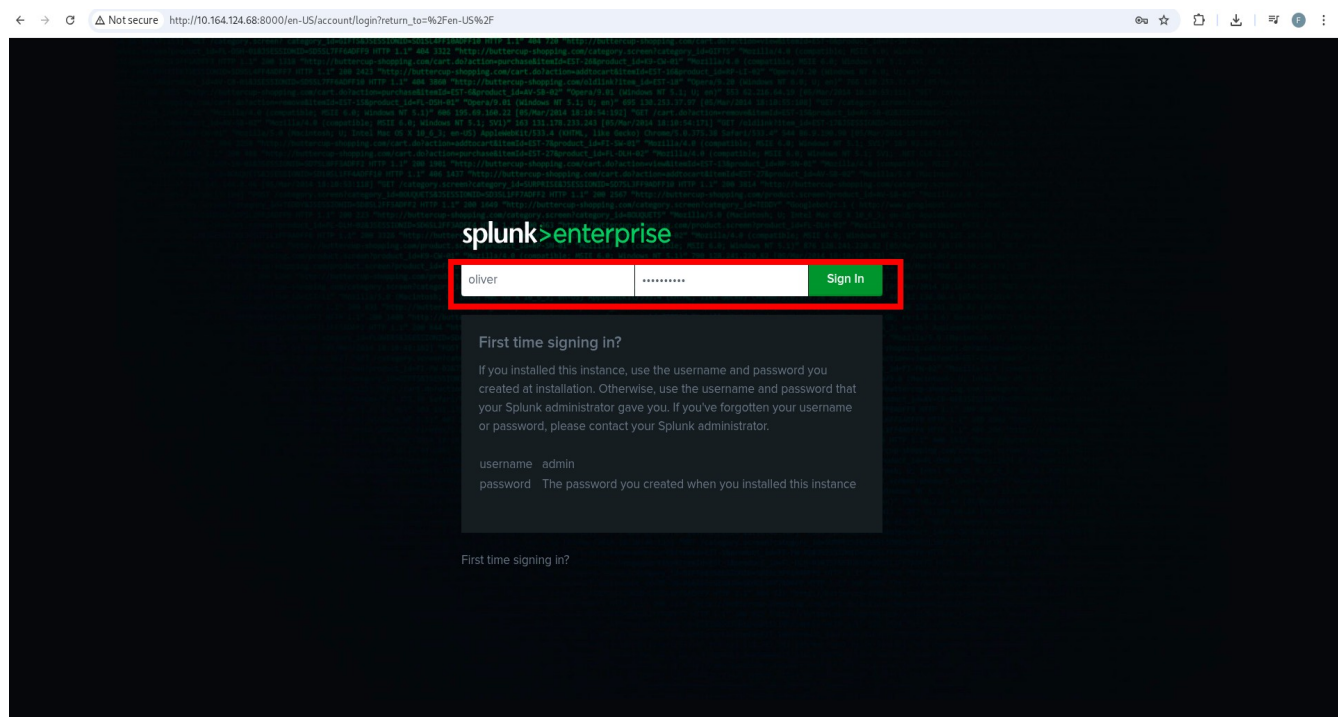


```
[WinEventLog://Microsoft-Windows-Sysmon/Operational]
disabled = 0
renderXml = true
index = main
sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational
```

B. Splunk Enterprise Setup

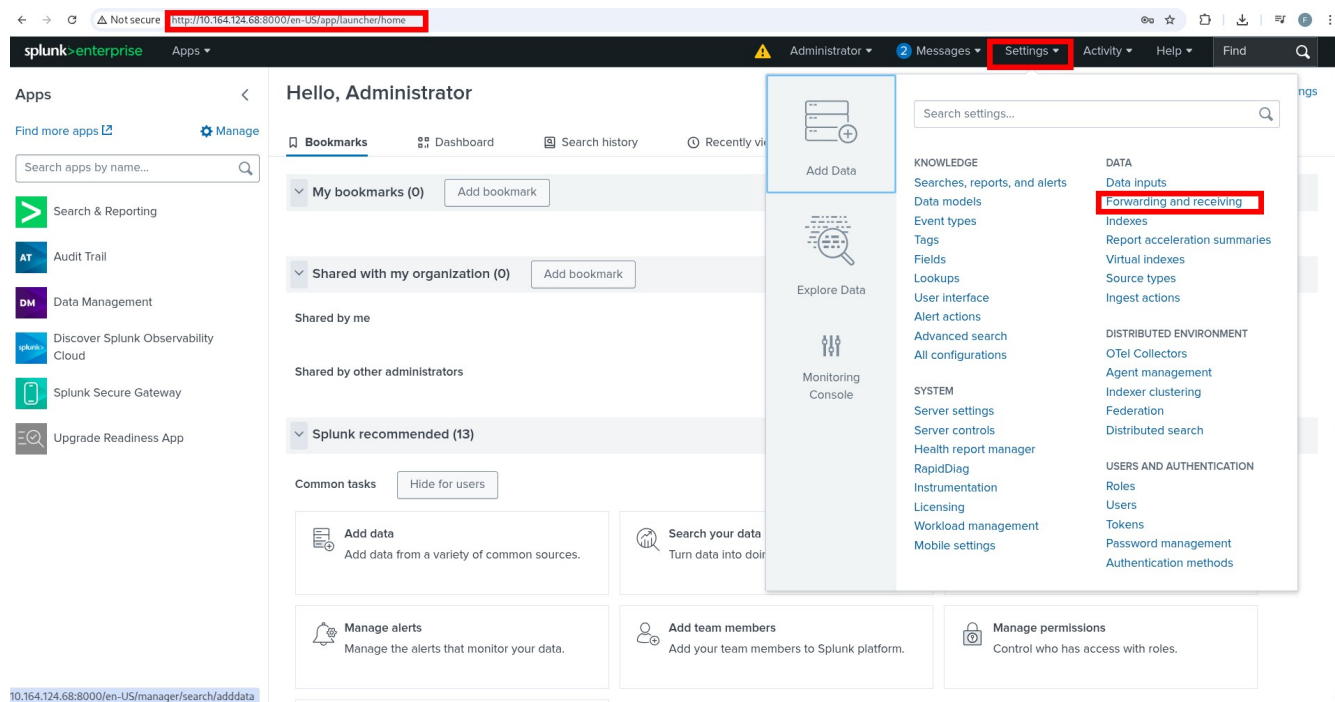
Step 1:

Insert splunk localhost/IP:8000 on address bar, enter login credentials and sign in



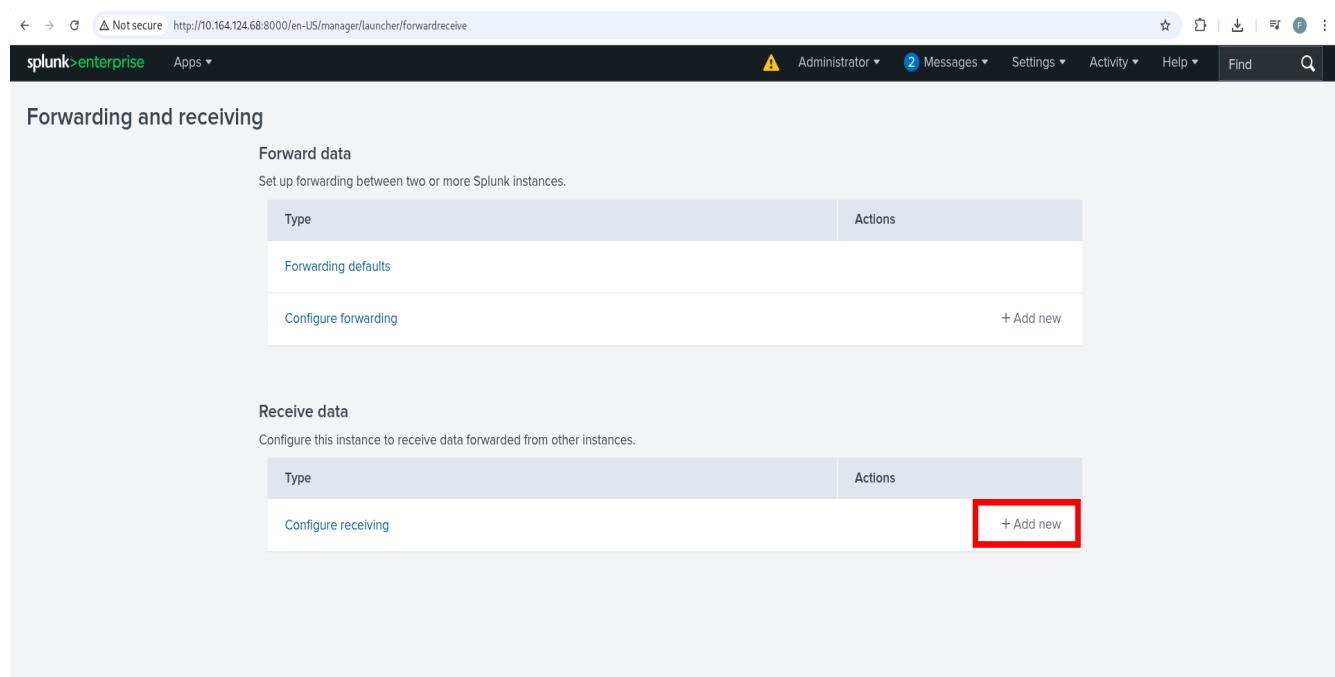
Step 2:

Click on “Settings” and choose “Forwarding and Receiving” option



Step 3:

Click on “Add New” to configure receiving of data



Step 4: Enter Port “9997” and save

← → ↻ Not secure http://10.164.124.68:8000/en-US/manager/launcher/data/inputs/tcp/cooked/_new?action=edit

splunk>enterprise Apps ▾ ⚠ Administrator ▾ 2 Messages ▾ Settings ▾ Activity ▾ Help ▾ Find 🔍

Add new
Forwarding and receiving ▸ Receive data ▸ Add new

Configure receiving
Set up this Splunk instance to receive data from forwarder(s).

Listen on this port

For example, 9997 will receive data on TCP port 9997.

Cancel Save

Step 5: Splunk Enterprise instance enabled to receive data from “UF”

← → ↻ Not secure http://10.164.124.68:8000/en-US/manager/launcher/data/inputs/tcp/cooked?msgid=3279600.312675810119453&ns=launcher&redirecting=true

splunk>enterprise Apps ▾ ⚠ Administrator ▾ 2 Messages ▾ Settings ▾ Activity ▾ Help ▾ Find 🔍

Receive data New Receiving Port
Forwarding and receiving ▸ Receive data

Successfully saved "9997".
Showing 1-1 of 1 item

filter 🔍 25 per page ▾

Listen on this port ▾	Status ▾	Actions
9997	Enabled Disable	Delete

C. Data Onboarding:

Step 1: Click on “Searching and Reporting”

← → ↻ Not secure http://10.164.124.68:8000/en-US/app/launcher/home

splunk>enterprise Apps ▾ ⚠ Administrator ▾ 2 Messages ▾ Settings ▾ Activity ▾ Help ▾ Find 🔍

Apps < Find more apps 🔍 Manage ⚙️
Search apps by name... 🔍

- Search & Reporting**
- AT Audit Trail
- DM Data Management
- Discover Splunk Observability Cloud
- Splunk Secure Gateway
- Upgrade Readiness App

Hello, Administrator Home page settings ⚙️

Bookmarks Dashboard Search history Recently viewed Created by you Shared with you

My bookmarks (0) Add bookmark

Shared with my organization (0) Add bookmark

Shared by me

Shared by other administrators

Splunk recommended (13)

Common tasks Hide for users

Add data
Add data from a variety of common sources.

Search your data
Turn data into doing with Splunk search.

Visualize your data
Create dashboards that work for your data.

Manage alerts
Manage the alerts that monitor your data.

Add team members
Add your team members to Splunk platform.

Manage permissions
Control who has access with roles.

Scenerio 1:

WinEventLog:Security

SPL Query: index=main host="WINDOWS001"

Note: Smart Mode

The screenshot shows the Splunk Enterprise interface. The search bar contains the query `index=main host="WINDOWS001"`. The search results show 99,263 events. The interface is in Smart Mode. The search results are displayed in a table format with columns for Time and Event. The first event is from 01/23/2026 02:49:19.317 PM, LogName=Security, EventCode=4688, EventType=0, ComputerName=WINDOWS-001. The second event is from 01/23/2026 02:49:19.117 PM, LogName=Security, EventCode=4798, EventType=0, ComputerName=WINDOWS-001. The third event is from 01/23/2026 02:49:19.116 PM, LogName=Security, EventCode=4798, EventType=0, ComputerName=WINDOWS-001.

Scenerio 2:

WinEventLog:Security

SPL Query: index=main sourcetype="WinEventLog:Security" EventCode=4625

| table _time Computer Account_Name Account_Domain Logon_Type Failure_Reason

| sort _time

The screenshot shows the Splunk Enterprise interface. The search bar contains the query `index=main sourcetype="WinEventLog:Security" EventCode=4625 | table _time Computer Account_Name Account_Domain Logon_Type Failure_Reason | sort _time`. The search results show 14 events. The interface is in Smart Mode. The search results are displayed in a table format with columns for _time, Computer, Account_Name, Account_Domain, Logon_Type, and Failure_Reason. The first event is from 2026-01-23 13:42:15.936, Computer=WINDOWS-001\$ugwoke, Account_Name=WORKGROUP, Account_Domain=WINDOWS-001, Logon_Type=2, Failure_Reason=Unknown user name or bad password. The second event is from 2026-01-23 13:42:18.851, Computer=WINDOWS-001\$ugwoke, Account_Name=WORKGROUP, Account_Domain=WINDOWS-001, Logon_Type=2, Failure_Reason=Unknown user name or bad password. The third event is from 2026-01-23 13:42:21.280, Computer=WINDOWS-001\$ugwoke, Account_Name=WORKGROUP, Account_Domain=WINDOWS-001, Logon_Type=2, Failure_Reason=Unknown user name or bad password. The fourth event is from 2026-01-23 13:42:24.366, Computer=WINDOWS-001\$ugwoke, Account_Name=WORKGROUP, Account_Domain=WINDOWS-001, Logon_Type=2, Failure_Reason=Unknown user name or bad password. The fifth event is from 2026-01-23 13:42:26.572, Computer=WINDOWS-001\$ugwoke, Account_Name=WORKGROUP, Account_Domain=WINDOWS-001, Logon_Type=2, Failure_Reason=Unknown user name or bad password. The sixth event is from 2026-01-23 13:42:28.900, Computer=WINDOWS-001\$ugwoke, Account_Name=WORKGROUP, Account_Domain=WINDOWS-001, Logon_Type=2, Failure_Reason=Unknown user name or bad password. The seventh event is from 2026-01-23 13:45:10.868, Computer=WINDOWS-001\$ugwoke, Account_Name=WORKGROUP, Account_Domain=WINDOWS-001, Logon_Type=2, Failure_Reason=Unknown user name or bad password. The eighth event is from 2026-01-23 13:45:21.786, Computer=WINDOWS-001\$ugwoke, Account_Name=WORKGROUP, Account_Domain=WINDOWS-001, Logon_Type=2, Failure_Reason=Unknown user name or bad password.

Scenerio 3:

WinEventLog:Security

SPL Query: index=main host="WINDOWS001" | stats count by ComputerName EventCode EventType Logon_ID SourceName

splunk>enterprise Apps Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

index=main host="WINDOWS001" | stats count by ComputerName EventCode EventType Logon_ID SourceName Time range: All time

183,483 events (before 1/25/26 11:37:16.000 AM) No Event Sampling

Events Patterns **Statistics (354)** Visualization

Show: 20 Per Page Format Preview: On

ComputerName	EventCode	EventType	Logon_ID	SourceName	count
WINDOWS-001	4611	0	0x3E7	Microsoft Windows security auditing.	144
WINDOWS-001	4616	0	0x3E5	Microsoft Windows security auditing.	1
WINDOWS-001	4624	0	0x0	Microsoft Windows security auditing.	5
WINDOWS-001	4624	0	0x13502BE	Microsoft Windows security auditing.	1
WINDOWS-001	4624	0	0x135031F	Microsoft Windows security auditing.	1
WINDOWS-001	4624	0	0x1350A1F	Microsoft Windows security auditing.	1
WINDOWS-001	4624	0	0x1350B90	Microsoft Windows security auditing.	1
WINDOWS-001	4624	0	0x15D6D1A	Microsoft Windows security auditing.	1
WINDOWS-001	4624	0	0x15D727C	Microsoft Windows security auditing.	1
WINDOWS-001	4624	0	0x15D729B	Microsoft Windows security auditing.	1
WINDOWS-001	4624	0	0x15EBE7F	Microsoft Windows security auditing.	1
WINDOWS-001	4624	0	0x15EBED6	Microsoft Windows security auditing.	1
WINDOWS-001	4624	0	0x15EC658	Microsoft Windows security auditing.	1
WINDOWS-001	4624	0	0x15EC6E8	Microsoft Windows security auditing.	1

Scenerio 4:

WinEventLog:Sysmon

SPL Query: index=main source="WinEventLog:Microsoft-Windows-Sysmon/Operational"

splunk>enterprise Apps Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

index=main source="WinEventLog:Microsoft-Windows-Sysmon/Operational" Time range: Last 24 hours

6,019 events (1/24/26 100:00:00 AM to 1/25/26 1:17:08.000 AM) No Event Sampling

Events (6,019) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 hour per column

Format Show: 20 Per Page View: List

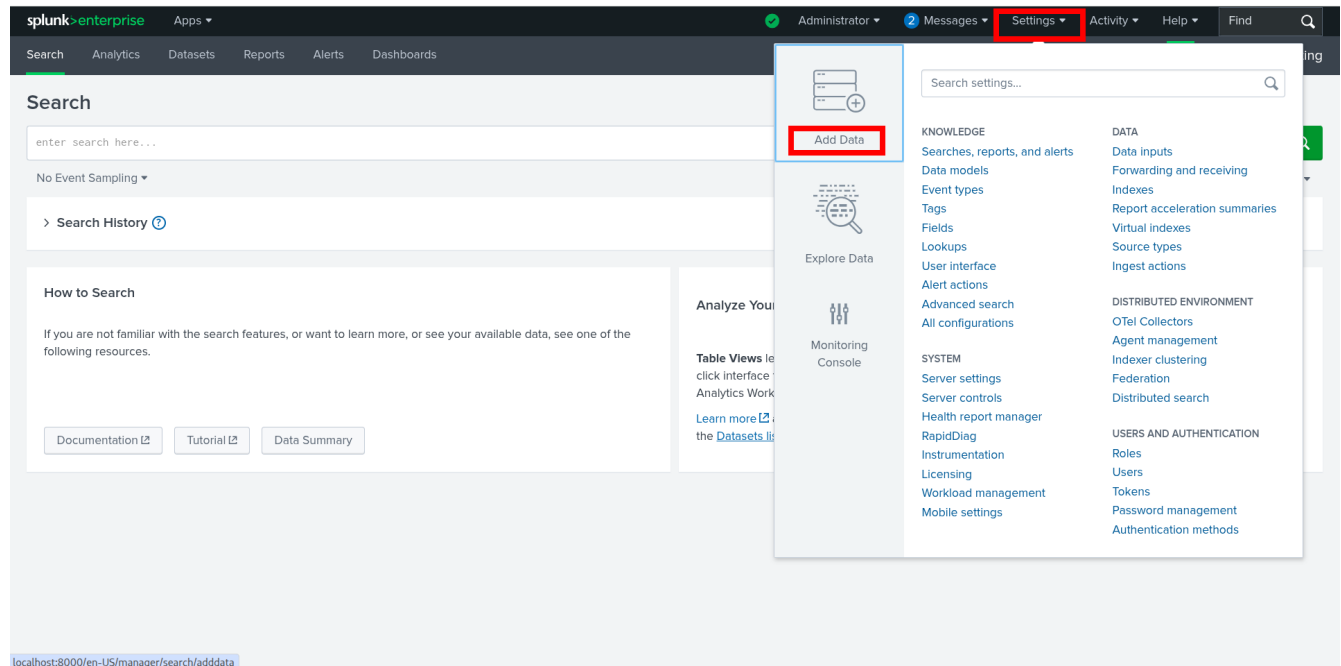
Time	Event
1/25/26 1:16:43.542 AM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffb9}' /><EventID>5</EventID><Version>3</Version><Level>4</Level><Task>5</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-01-25T00:16:43.542Z' /><EventRecordID>6019</EventRecordID><Correlation><Execution ProcessID='8368' ThreadID='7764' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WINDOWS-001</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'></Data><Data Name='UtcTime'>2026-01-25 00:16:43.542</Data><Data Name='ProcessGuid'>(7560cac2-60eb-6975-6b0e-000000003b00)</Data><Data Name='ProcessID'>10724</Data><Data Name='Image'>C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe</Data><Data Name='User'>NT AUTHORITY\SYSTEM</Data></EventData></Event> host = WINDOWS-001 source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = xmlwineventlog
1/25/26 1:16:43.431 AM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffb9}' /><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-01-25T00:16:43.431Z' /><EventRecordID>6018</EventRecordID><Correlation><Execution ProcessID='8368' ThreadID='7764' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>WINDOWS-001</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'></Data><Data Name='UtcTime'>2026-01-25 00:16:43.428</Data><Data Name='ProcessGuid'>(7560cac2-60eb-6975-6b0e-000000003b00)</Data><Data Name='ProcessID'>10724</Data><Data Name='Image'>C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe</Data><Data Name='FileVersion'>10.2.0</Data><Data Name='Description'>Registry monitor</Data><Data Name='Product'>Splunk Application</Data><Data Name='Company'>Splunk Inc.</Data><Data Name='OriginalFileName'>splunk-regmon.exe</Data></EventData></Event>

C. Log Ingestion

Uploading DNS log file into Splunk Enterprise

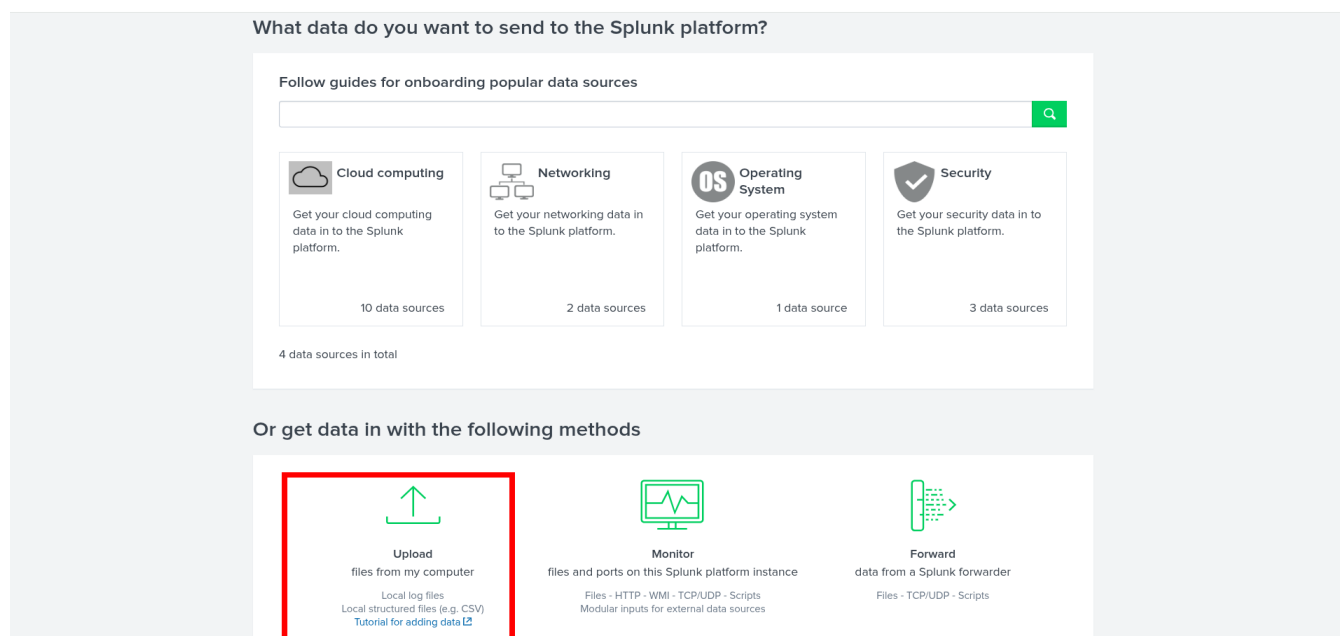
Step 1:

Click on “Settings” and then click on “Add data”



Step 2:

Click on “Upload file from my computer”



Step 3:
Click on “Select file”

The screenshot shows the Splunk Enterprise interface for the 'Add Data' process. The top navigation bar includes the Splunk logo, 'Apps', and user roles like 'Administrator'. A progress bar indicates the current step is 'Select Source'. The main content area is titled 'Select Source' and instructs the user to choose a file to upload. Below this, it says 'Selected File: No file selected'. A button labeled 'Select File' is highlighted with a red rectangular box. Below the button is a large empty box for dropping a file, with the text 'Drop your data file here' and 'The maximum file upload size is 500 Mb'. At the bottom, there is an 'FAQ' section with three questions.

Step 4:
Click “Next” after file upload

This screenshot shows the same 'Select Source' page after a file has been uploaded. The 'Selected File' now displays 'dns.log.gz'. The 'Select File' button is still present but no longer highlighted. A green checkmark icon and the text 'File Successfully Uploaded' are displayed at the bottom of the file drop area, enclosed in a red rectangular box. The 'Next >' button in the top progress bar is also highlighted with a red rectangular box. The rest of the page, including the progress bar and FAQ section, remains the same as in the previous screenshot.

Click “Save As” to save the log file”

splunk>enterprise Apps ▾

[Administrator](#) ▾ |
 [Messages](#) |
 [Settings](#) ▾ |
 [Activity](#) ▾ |
 [Help](#) ▾ |
 Find

Add Data

Select Source — ● — ● — Set Source Type — Input Settings — Review — Done

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: **dns.log.gz** [View Event Summary](#)

Save As

Format ▾	Select... ▾	Select... ▾	< Prev	1	2	3	4	5	6	7	8	...	Next >
Event Breaks													
Timestamp													
Advanced													
1	▲	1/25/26 12:08:44.000 PM	1331901005.510000 0\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00 F F F F 1 - - F	CWGtK431H9XuaTn4f1	192.168.202.2100	45658	192.168.27.203	137	udp	33008	*\x00\x0		
			timestamp = none										
2	▲	1/25/26 12:08:44.000 PM	1331901015.070000 1 C_INTERNET	C36a282J1jz7BsbGH	192.168.202.76	137	192.168.202.255	137	udp	57402	HPE8AA67		
			timestamp = none										
3	▲	1/25/26 12:08:44.000 PM	1331901015.820000 1 C_INTERNET	C36a282J1jz7BsbGH	192.168.202.76	137	192.168.202.255	137	udp	57402	HPE8AA67		
			timestamp = none										
4	▲	1/25/26 12:08:44.000 PM	1331901016.570000 1 C_INTERNET	C36a282J1jz7BsbGH	192.168.202.76	137	192.168.202.255	137	udp	57402	HPE8AA67		
			timestamp = none										
5	▲	1/25/26 12:08:44.000 PM	1331901005.860000 1 C_INTERNET	C36a282J1jz7BsbGH	192.168.202.76	137	192.168.202.255	137	udp	57398	WPAD		
			timestamp = none										
6	▲	1/25/26	1331901006.610000	C36a282J1jz7BsbGH	192.168.202.76	137	192.168.202.255	137	udp	57398	WPAD		

Enter file name, select as appropriate and save

The screenshot shows the Splunk web interface. A modal dialog titled "Set Source Type" is open in the center. The dialog has the following fields:

- Name:** A text input field containing "DNSLOG".
- Description:** A text input field containing "My DNS log file".
- Category:** A dropdown menu with "Network & Security" selected.
- App:** A dropdown menu with "Search & Reporting" selected.

At the bottom of the dialog are two buttons: "Cancel" and "Save". The "Save" button is highlighted with a red rectangular box. In the background, the "Add Data" page is visible, showing a list of source types on the left and a table of event data on the right. The table has columns for event number, timestamp, source type, and various fields like IP addresses and ports.

Step 7:
Click “Review”

splunk>enterprise Apps

Administrator Messages Settings Activity Help Find

Add Data

Select Source Set Source Type Input Settings Review Done

< Back Review >

Input Settings

Optionally set additional input parameters for this data input as follows:

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

☒ Constant value
☐ Regular expression on path
☐ Segment in path

Host field value:

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index: [Create a new index](#)

FAQ

- > How do indexes work?
- > How do I know when to create or use multiple indexes?

Step 8:
Review and submit

splunk>enterprise Apps

Administrator Messages Settings Activity Help Find

Add Data

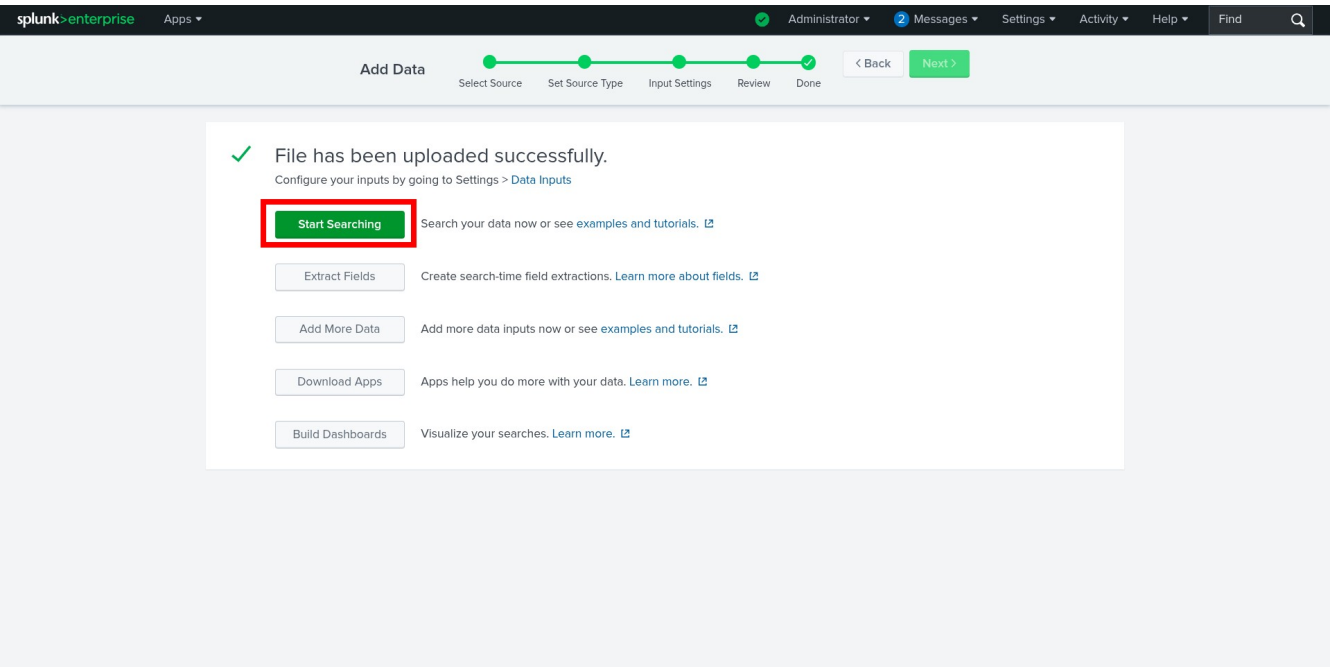
Select Source Set Source Type Input Settings Review Done

< Back Submit >

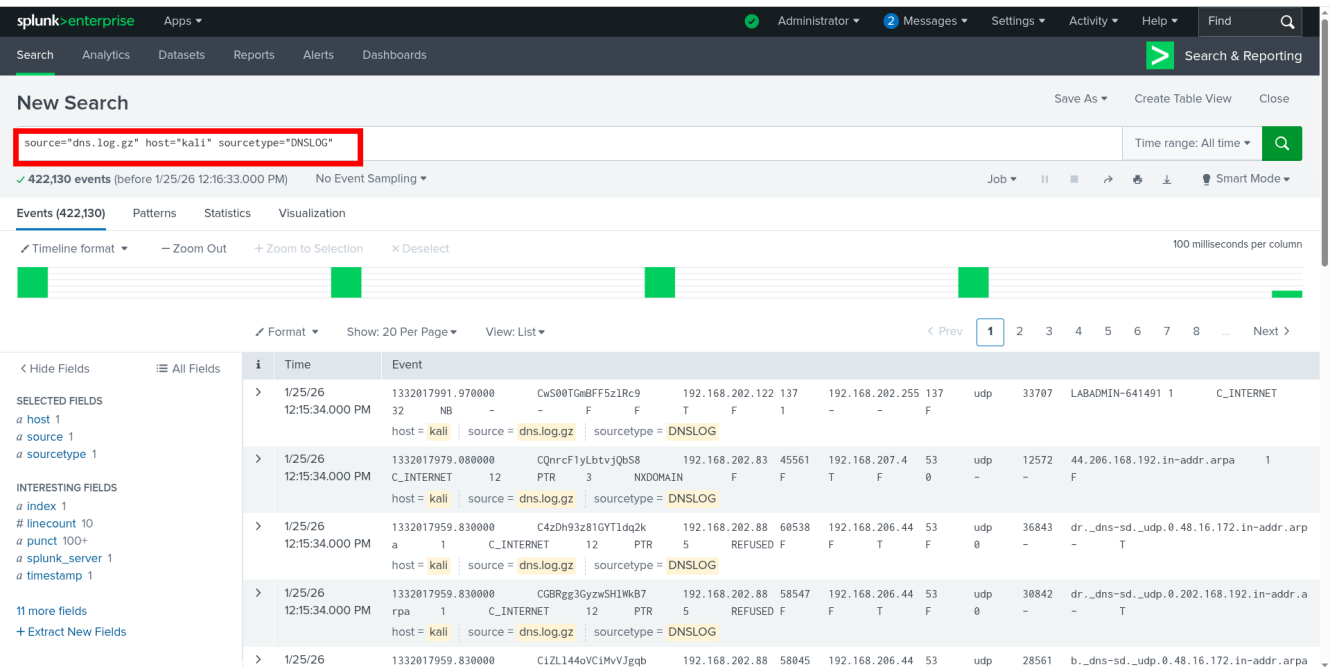
Review

Input Type Uploaded File
File Name dns.log.gz
Source Type DNSLOG
Host kali
Index Default

Step 9:
Click on “Start searching” to go to the search page



Step 10:
DNS log successfully parsed



New File Extraction:

Step 1: Click on “Extract New File”

splunk>enterprise

Apps

Administrator

Messages

Settings

Activity

Help

Find

Search

Analytics

Datasets

Reports

Alerts

Dashboards

Search & Reporting

New Search

Save As

Create Table View

Close

source="dns.log.gz" host="kali" sourcetype="DNSLOG"

Time range: All time

422,130 events (before 1/25/26 12:16:33.000 PM)

No Event Sampling

Job

Smart Mode

Events (422,130)

Patterns

Statistics

Visualization

Timeline format

Zoom Out

Zoom to Selection

Deselect

100 milliseconds per column

Format

Show: 20 Per Page

View: List

Prev

1

2

3

4

5

6

7

8

Next

< Hide Fields

All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

INTERESTING FIELDS

a index 1

linecount 10

a punct 100+

a splunk_server 1

a timestamp 1

11 more fields

Extract New Fields

i

Time

Event

>

1/25/26 12:15:34.000 PM

1332017991.970000

Cw500Tgm8FF5z1Rc9

192.168.202.122 137

192.168.202.255 137

udp

33707

LABADMIN-641491 1

C_INTERNET

>

1/25/26 12:15:34.000 PM

1332017979.080000

CQnrcFtYlbtvjQbS8

192.168.202.83 45561

192.168.207.4 53

udp

12572

44.206.168.192.in-addr.arpa 1

C_INTERNET

>

1/25/26 12:15:34.000 PM

1332017959.830000

C4zDh93z81GYTldq2k

192.168.202.88 60538

192.168.206.44 53

udp

36843

dr._dns-sd._udp.0.48.16.172.in-addr.arpa

C_INTERNET

>

1/25/26 12:15:34.000 PM

1332017959.830000

CGBRgg3GyzwSH1WkB7

192.168.202.88 58547

192.168.206.44 53

udp

30842

dr._dns-sd._udp.0.202.168.192.in-addr.arpa

C_INTERNET

>

1/25/26 12:15:34.000 PM

1332017959.830000

CiZL144oVCiMvVJgqb

192.168.202.88 58045

192.168.206.44 53

udp

28561

b._dns-sd._udp.0.48.16.172.in-addr.arpa

C_INTERNET

Step 2: Select any of the events

splunk>enterprise

Apps

Administrator

Messages

Settings

Activity

Help

Find

Extract Fields

Select Sample

Select Method

Select Fields

Save

Next

Existing fields

Select Sample Event

Choose a source or source type, select a sample event, and click Next to go to the next step. The field extractor will use the event to extract fields. [Learn more](#)

I prefer to write the regular expression myself

Source type
DNSLOG

Time Range
Last 90 days

Events

1,000 events (10/27/25 12:00:00.000 AM to 1/25/26 12:19:01.000 PM)

20 per page

Prev

1

2

3

4

5

6

7

8

Next

filter

Apply

Sample: 1,000 events

All events

_raw

1332017991.970000

Cw500Tgm8FF5z1Rc9

192.168.202.122 137

192.168.202.255 137

udp

33707

LABADMIN-641491 1

C_INTERNET

32

NB

-

-

F

F

T

1332017979.080000

CQnrcFtYlbtvjQbS8

192.168.202.83 45561

192.168.207.4 53

udp

12572

44.206.168.192.in-addr.arpa 1

C_INTERNET

12

PTR

3

NXDOMAIN

1332017959.830000

C4zDh93z81GYTldq2k

192.168.202.88 60538

192.168.206.44 53

udp

36843

dr._dns-sd._udp.0.48.16.172.in-addr.arpa

1

C_INTERNET

12

PTR

5

Step 5:
Highlight the source IP to give it a name and click “Add Extraction”

Extract Fields

Select Sample Select Method Select Fields Validate Save < Back Next > Existing fields >

Select Fields

Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlighted values in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. [Learn more](#)

1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET 12 PTR 3 NXDOMAIN F T F 0 - - F

Extract Require

Field Name src_ip

Sample Value 192.168.202.83

Add Extraction

Step 6:
Highlight the source port to give it a name and click “Add Extraction”

Extract Fields

Select Sample Select Method Select Fields Validate Save < Back Next > Existing fields >

Select Fields

Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlighted values in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. [Learn more](#)

1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET 12 PTR 3 NXDOMAIN F T F 0 - - F

Show Regular Expression > Extract Require View in Search &

Field Name src_port

Sample Value 45561

Add Extraction

Preview

If you see incorrect results below, click an additional value to highlight it. Highlight its values to improve the extraction. You can remove incorrect values in the next step.

Events src_ip

1,000 events (10/27/25 12:00:00.000 AM to 1/25/26 12:25:18.000 PM) 20 per page < Prev 1 2 3 4 5 6 7 8 ... Next >

filter Apply Sample: 1,000 events All events All Events Matches Non-Matches

_raw	src_ip
1332017991.970000 - F F CwS00TCm6FF5z1Rc9 T F 1 192.168.202.122 137 192.168.202.255 137 udp 33707 LABADMIN-641491 1 C_INTERNET 32 NB - 192.168.202.122	192.168.202.122
1332017979.080000 3 NXDOMAIN CQnrcF1yLbtvjQbS8 F F T 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET 12 PTR 192.168.202.83	192.168.202.83

Step 7:

Highlight the destination IP to give it a name and click “Add Extraction”

Select Fields

Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlighted values in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. [Learn more](#)

Sample Event: 1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET 12 PTR 3 NXDOMAIN F T F 0 - - F

Field Name: **dst_ip**

Sample Value: 192.168.207.4

Add Extraction

Events: src_ip src_port

1,000 events (10/27/25 12:00:00.000 AM to 1/25/26 12:26:24.000 PM)

	_raw	src_ip	src_port
✓	1332017991.970000 - - F Cw500TgmBFF5z1Rc9 192.168.202.122 137 192.168.202.255 137 udp 33707 LABADMIN-641491 1 C_INTERNET 32 NB 192.168.202.122 137	192.168.202.122	137
✓	1332017979.080000 PTR 3 NXDOMAIN F F T F 0 - - F 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET 12 192.168.202.83 45561	192.168.202.83	45561

Step 8:

Highlight the destination port to give it a name, click “Add Extraction” and then “Next”

Let us stop with these four fields

Select Fields

Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlighted values in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. [Learn more](#)

Sample Event: 1332017979.080000 CQnrcF1yLbtvjQbS8 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 C_INTERNET 12 PTR 3 NXDOMAIN F T F 0 - - F

Field Name: **dst_port**

Sample Value: 53

Add Extraction

Events: src_ip src_port **dst_ip**

1,000 events (10/27/25 12:00:00.000 AM to 1/25/26 12:27:34.000 PM)

	_raw	src_ip	src_port	dst_ip
✓	1332017991.970000 C_INTERNET 32 NB - - F 192.168.202.122 137 192.168.202.255 137 udp 33707 LABADMIN-641491 1 192.168.202.122 137 192.168.202.255	192.168.202.122	137	192.168.202.255
✓	1332017979.080000 C_INTERNET 12 PTR 3 NXDOMAIN F F T F 0 - - F 192.168.202.83 45561 192.168.207.4 53 udp 12572 44.206.168.192.in-addr.arpa 1 192.168.202.83 45561 192.168.207.4	192.168.202.83	45561	192.168.207.4

Step 9: View the created fields and click “Next”

Extract Fields | Select Sample | Select Method | Select Fields | **Validate** | Save | < Back | **Next >** | Existing fields >

Validate

Validate your field extractions and remove values that are incorrectly highlighted in the Events tab. In the field tabs, inspect the extracted values for each field, and optionally click a value to apply it as a search filter to the Events tab event list.

Show Regular Expression > | View in Search >

Events | src_ip | src_port | dst_ip | dst_port

✓ 1,000 events (10/27/25 12:00:00.000 AM to 1/25/26 12:28:46.000 PM) | 20 per page | < Prev | **1** | 2 | 3 | 4 | 5 | 6 | 7 | 8 | ... | Next >

filter [] Apply | Sample: 1,000 events | All events | All Events | Matches | Non-Matches

_raw	src_ip	src_port	dst_ip	dst_port
✓ 1332017991.970000 CwS00TcmBFF5z1Rc9 641491 1 C_INTERNET 32 NB - - F F T F 1 - - udp 33707 LABADMIN- 192.168.202.122 137 192.168.202.255 137	192.168.202.122	137	192.168.202.255	137
✓ 1332017979.080000 CQnrcF1yLbtvjQbS8 44.206.168.192.192.in-addr.arpa 1 C_INTERNET 12 PTR 3 NXDOMAIN F F T F 0 - udp 12572 192.168.202.83 45561 192.168.207.4 53	192.168.202.83	45561	192.168.207.4	53
✓ 1332017959.830000 C4zDh93z81GYT1dq2k dr._dns-sd._udp.0.48.16.172.in-addr.arpa - - T 1 C_INTERNET 12 PTR 5 REFUSED F F T F 0 - udp 36843 192.168.202.88 60538 192.168.206.44 53	192.168.202.88	60538	192.168.206.44	53
✓ 1332017959.830000 CGBRgg3GyzwSH1Wk87 dr._dns-sd._udp.0.202.168.192.in-addr.arpa 1 C_INTERNET 12 PTR 5 REFUSED F F T F 0 - udp 30842 192.168.202.88 58547 192.168.206.44 53	192.168.202.88	58547	192.168.206.44	53

Step 10: Click “Finish” to complete the extraction

Extract Fields | Select Sample | Select Method | Select Fields | Validate | **Save** | < Back | **Finish >**

Save

Name the extraction and set permissions.

Extractions Name **EXTRACT src_ip,src_port,dst_ip,dst_port**

Owner oliver

App search

Permissions Owner App All apps

Source type DNSLOG

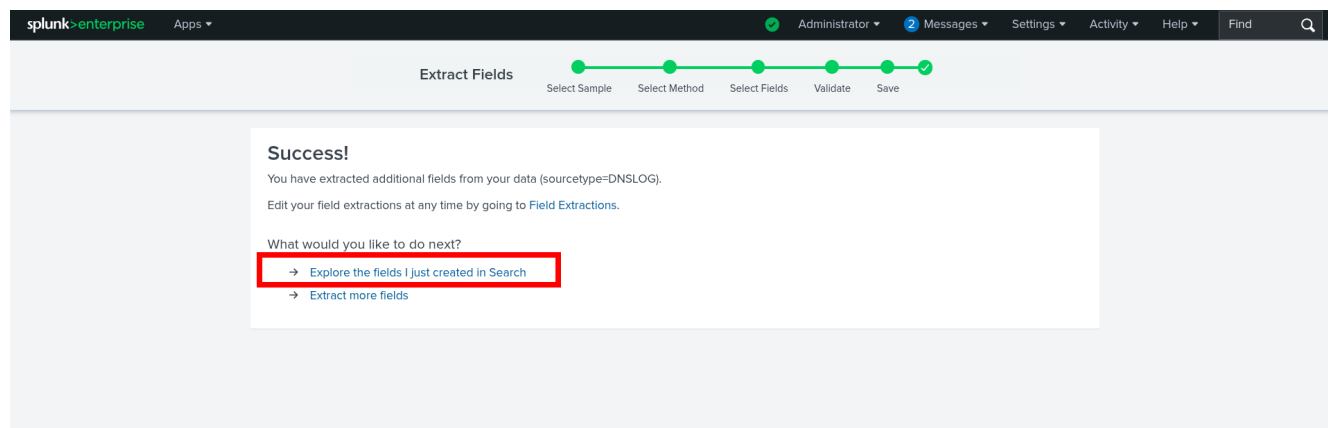
Sample event 1332017979.080000 CQnrcF1yLbtvjQbS8 12572 44.206.168.192.192.in-addr.arpa 1 C_INTERNET 12 PTR 3 NXDOMAIN F

Fields src_ip,src_port,dst_ip,dst_port

Regular Expression ^(?!(?:\n|\t))*([2])(?P<src_ip>[^\t]+)\t(?P<src_port>[^\t]+)\t(?P<dst_ip>[^\t]+)\t(?P<dst_port>[^\t]+)

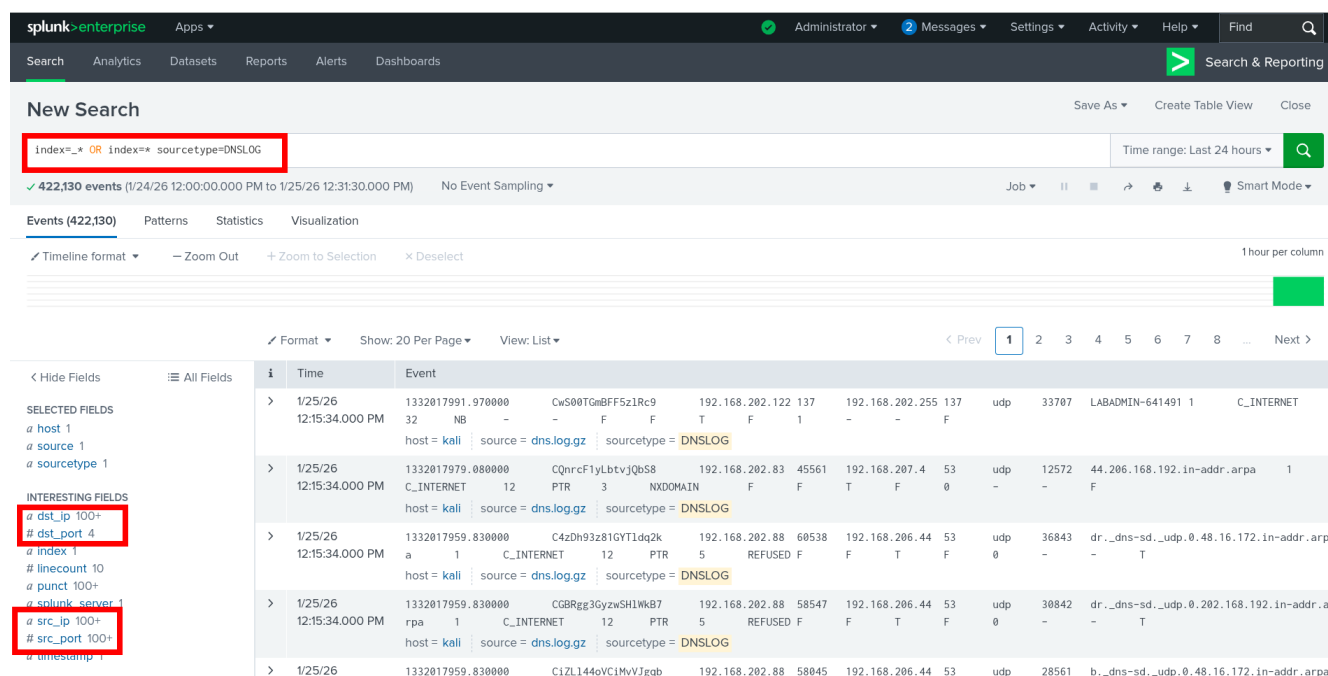
Step 11:

Click to explore the newly created fields



Step 12:

Newly created field now found on the left sidebar



Step 13:

Search:
index=* sourcetype=DNSLOG | stats count by src_ip src_port dst_ip dst_port

splunkenterpriseAppsAdministrator2 MessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboardsSearch & Reporting

New SearchSave AsCreate Table ViewClose

index=* OR index=* sourcetype=DNSLOGstats count by src_ip src_port dst_ip dst_portTime range: Last 24 hours

422,130 events (1/24/26 12:00:00.000 PM to 1/25/26 12:33:22.000 PM)No Event SamplingJobSmart Mode

EventsPatternsStatistics (112,771)Visualization

Show: 20 Per PageFormatPreview: On< Prev12345678...Next>

src_ip	src_port	dst_ip	dst_port	count
10.10.10.10	137	10.255.255.255	137	69
10.10.117.209	53291	192.168.207.4	53	4
10.10.117.209	53292	192.168.207.4	53	3
10.10.117.209	53293	192.168.207.4	53	1
10.10.117.209	53294	192.168.207.4	53	5
10.10.117.209	53295	192.168.207.4	53	7
10.10.117.209	53296	192.168.207.4	53	10
10.10.117.209	53297	192.168.207.4	53	7
10.10.117.209	53298	192.168.207.4	53	5
10.10.117.209	53299	192.168.207.4	53	5
10.10.117.209	53300	192.168.207.4	53	7
10.10.117.209	53301	192.168.207.4	53	7
10.10.117.209	53302	192.168.207.4	53	4
10.10.117.209	53303	192.168.207.4	53	6